1. Mixed Methods Design and Reporting

How can we explain differences in EFL students' essay revisions after teacher feedback?

DEDC02 · Advanced Research Method

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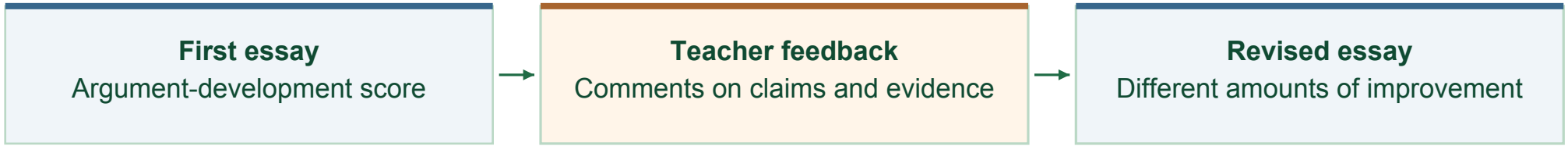
2. Today’s learning aims

By the end of this lesson, you will be able to:

- **Locate mixed methods** within the wider methods landscape and explain why a problem needs both kinds of evidence.
- **Choose and diagram** a convergent, explanatory sequential, or exploratory sequential design.
- **Justify sampling** choices, sample relationships, timing, and sample sizes across the two strands.
- **Use a joint display** to develop an integrated explanation and state its limits.
- **Report and appraise** the design, integration, and evidence supporting a study’s conclusions.

3. Feedback was given. What changed?

In an EFL writing class, students receive comments on the argument in a first draft. Some develop their reasoning; others mainly correct grammar.



We need to understand both the pattern of revision gains and how learners used the feedback.

Running teaching example. All study procedures, numbers, and findings in this example are illustrative.

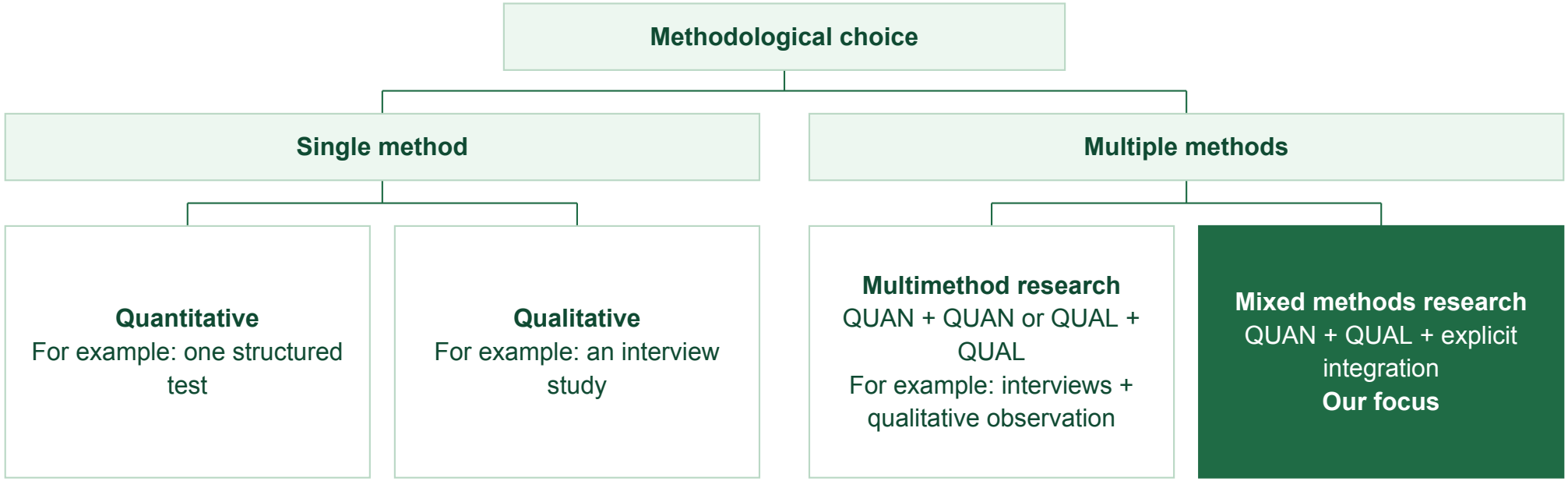
4. What must we find out?

Question	What evidence is needed?
QUAN: How much does argument development improve between first and revised essays?	Rubric scores and the distribution of revision gains.
QUAL: How do students interpret and use teacher feedback?	Interviews supported by the comments and revised drafts.
Mixed: How do students' explanations help us understand differences in revision gains?	An explicit link between score patterns and qualitative findings.

Mixed methods needs a reason for combining the strands and an answer developed through their relationship.

5. Where does mixed methods sit?

Would interviews plus classroom observation already make a study mixed methods?



The highlighted branch integrates quantitative and qualitative evidence to answer a shared research problem.

How are the terms used here?

This teaching map uses **multimethod** for several methods within the same quantitative or qualitative tradition. Authors use these labels differently; identify the evidence and its integration when reading a study.

6. Design choices around the mix

Choosing mixed methods answers one design question. What else must the study specify?

Decision	Place it in the whole study
Philosophy and theory	Justify the study's assumptions and theoretical lens. Mixed methods can be used with different philosophical positions.
Methodological choice	Single method, multimethod, or mixed methods: which kinds of evidence are needed?
Strategy and design	An experiment, case study, participatory project, or evaluation can contain a convergent, explanatory, or exploratory design.
Time and procedures	Cross-sectional or longitudinal inquiry; specify participants, instruments, analysis, and integration points.

Where do "mixed method" and "mixed model" fit?

The earlier W14 map distinguishes mixing at the methods/phases level (**mixed method**) from mixing across several stages (**mixed model**). We use **mixed methods** broadly here, and state exactly what is integrated, where, and why.

The three core designs can also form parts of embedded or multiphase studies. A design name is a starting point for specifying the whole study.

7. Three designs, three relationships

The same EFL feedback problem can be studied through three different relationships. Select a design.

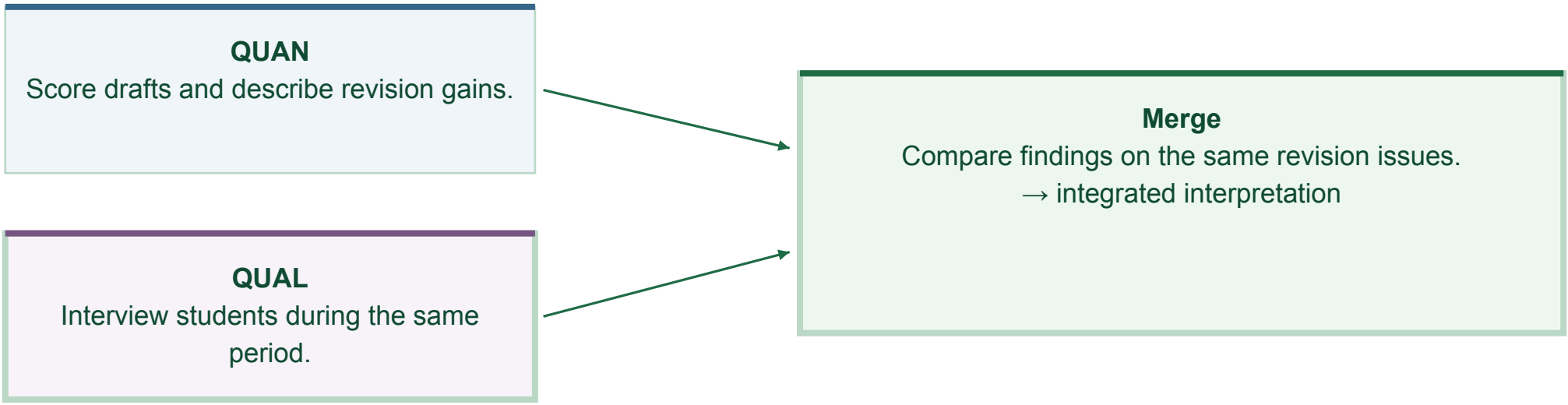
Design	Purpose	Sequence
Convergent	Compare results	QUAN + QUAL
Explanatory sequential	Explain results	QUAN → QUAL
Exploratory sequential	Develop a tool	QUAL → develop → QUAN

Compare, explain, develop: the integration purpose determines the relationship between the strands.

Convergent: compare two views of revision

How do score patterns and students' descriptions of using feedback compare?

Completed diagram



- Analyse each strand appropriately, then compare common concepts.
- Look for agreement, added detail, and differences.
- Concurrent collection alone does not demonstrate integration.

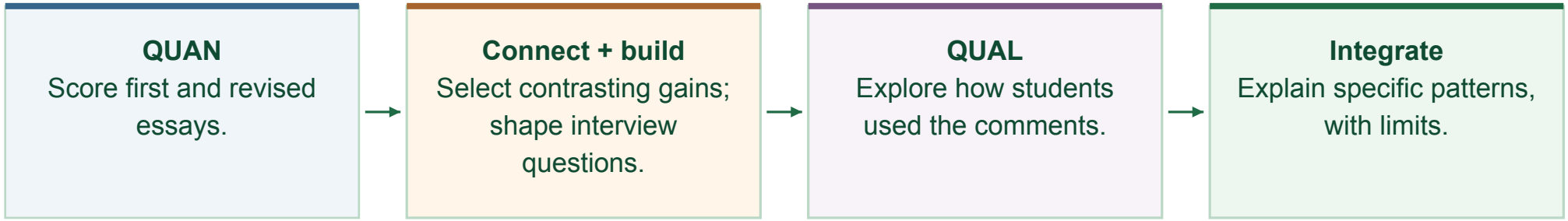
Plan a comparable comparison

Decision	EFL feedback example
Shared focus	Argument development, rather than mixing argument scores with unrelated attitudes.
Samples and time	Describe which students contribute to each strand and which writing task is examined.
Visible product	A joint display connecting revision results with interview findings.

Explanatory sequential: explain different revision gains

How do students' explanations help us understand the initial score patterns?

Completed diagram



- The initial results guide whom to interview and what to ask.
- Bring the linked findings together again when interpreting the results.

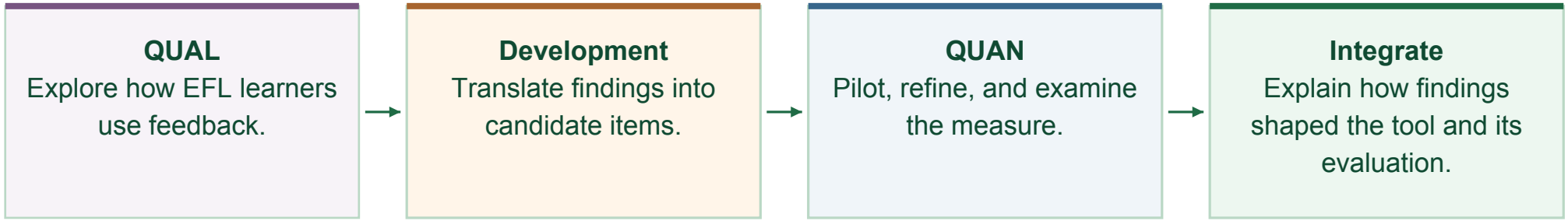
Inspect the bridge between phases

Initial result	Follow-up decision
Larger revision gains	Ask how comments were used to strengthen claims and evidence.
Little or no revision gain	Ask which comments were difficult to interpret or apply.
Unexpected case	Examine the feedback, draft changes, and student explanation together.

Exploratory sequential: develop a feedback-use measure

Which aspects of using teacher feedback should a new questionnaire measure?

Completed diagram



- Use when the relevant concepts or a suitable measure need development.
- The development phase must be visible. Interviewing first is not enough.

Trace findings into a measure

Qualitative finding	Candidate feature	Check
Students can identify a weakness but cannot plan a revision.	Separate feedback understanding from revision planning.	Content review and cognitive interviews.
Students struggle to use comments about supporting evidence.	Items about selecting evidence and linking it to a claim.	Pilot and evaluate score interpretations.

Additional design checks

How do we read these diagrams?

QUAN means quantitative; QUAL means qualitative. + shows concurrent strands; → shows sequence. Labels must explain what passes between phases. State relative emphasis separately.

How do the core designs fit a larger study?

Experiments, case studies, participatory projects, and evaluations can contain one or more core designs. Name the larger framework and explain the integration within it.

What else must the design specify?

Specify purpose, timing, emphasis, sample relationships, an integration product, and how the methods fit the study’s assumptions.

8. Which design fits the question?

Scenario 1 / 5

EFL essay scores show that some students improve much more than others after teacher feedback. Select students from the different gain patterns and interview them about their revisions.

Suggested design: Explanatory sequential. Explanatory sequential: the quantitative pattern guides interview sampling and questions, and the interviews help interpret that pattern.

Scenario 2 / 5

You want to develop an AI-feedback-literacy questionnaire for EFL learners. First explore learners' experiences, build candidate items, and then evaluate the questionnaire quantitatively.

Suggested design: Exploratory sequential. Exploratory sequential: qualitative findings build a measure, followed by development and quantitative evaluation.

Design practice - continued

Scenario 3 / 5

During the same writing unit, collect rubric scores and interviews about using teacher feedback. Analyse each strand, then compare findings on argument development.

Suggested design: Convergent. Convergent: compare the results around shared writing concepts. The comparison is the integration, not simply collecting both kinds of data.

Scenario 4 / 5

A peer-assessment study finds different speaking-score gains across EFL groups. Follow up the larger and smaller gain patterns with interviews to understand how feedback was used.

Suggested design: Explanatory sequential. Explanatory sequential: initial score differences determine the follow-up focus and sample.

Design practice - continued

Scenario 5 / 5

Before designing feedback in a new English-speaking practice app, explore learners' needs. Use those findings to develop feedback features and then test the features quantitatively.

Suggested design: Exploratory sequential. Exploratory sequential: the qualitative phase informs a feature-development phase before quantitative testing.

9. Sampling involves several decisions

Must both strands use the same people, the same numbers, and the same time period?

1. Selection	2. Relationship	3. Timing
How will cases enter each strand? Use probability, purposive, or a justified combination of strategies.	Do the strands study the same cases, a subset, separate samples, or linked levels?	Are samples planned concurrently, or does an earlier phase guide later selection?

First define the population and units: people, schools, drafts, events, or time points. Sample sizes and selection rules can differ across strands.

Start from the question

Comparing individual score changes with interview accounts requires linked individuals. Studying classroom conditions may instead require student evidence linked to teacher or school evidence. Specify the relationship needed for the intended inference.

10. How do the strand samples relate?

Identify who or what contributes to each strand before choosing a label.

Identical Same learners → scores + interviews Every sampled learner contributes to both strands. Individual accounts can be compared with that learner's results.	**Nested** 12 interviewees within 120 scored learners One strand uses a subset of the other sample. Our running study uses this relationship.
Parallel samples Different samples A and B; same population Different learners from the same target population contribute to each strand. Compare patterns without assuming individual matches.	**Multilevel** Learner scores ↔ teacher accounts Strands sample linked units at different levels or roles, such as learners and their teachers. Explain the link between them.

Nested samples and multistage selection

Selecting schools and then students describes stages of selection. For the relationship between strands, ask whether they examine the same units, a subset at the same level, or different linked levels. These are separate descriptions.

11. When are the samples selected?

Concurrent: sampling in the same period. Sequential: sampling across phases, with earlier findings able to guide the next phase.

Sample relationship	Concurrent example	Sequential example
Identical	The same learners provide scores and interviews during one period.	Interview the same learners after analysing their scores.
Nested	Interview a subset while the larger cohort completes the writing task.	Select interviewees from the scored cohort after examining gains.
Parallel samples	Score one sample; interview other learners from the same population.	Develop items through interviews; test them with another sample from that population.
Multilevel	Collect learner scores and linked teacher interviews in the same term.	Use learner patterns to select classrooms and interview their teachers.

“Parallel” has two uses

Parallel/concurrent sampling describes timing. **Parallel samples** describes different samples from the same population. State which meaning you intend.

A sequential design can recruit new participants. A concurrent design can use a nested subset.

12. How will you select cases?

A sample relationship does not tell us how its members were selected.

Probability sampling Use a sampling frame and known, non-zero selection chances. Example: randomly select learners from an eligible register to estimate an average.	**Purposive sampling** Select information-rich cases using study-relevant criteria, without probability selection. Example: seek learners with contrasting experiences of using feedback.
Stratified purposive sampling Identify relevant subgroups; select informative cases within each. Example: purposefully select larger-gain and limited-gain learners.	**Purposeful random sampling** Define an eligible pool purposefully, then select randomly within it. Example: randomly select interviewees from consenting limited-gain learners.

Convenience recruitment and the scope of claims

Recruiting an available class is convenience sampling. State the access limits and the population to which random selection applies. Later random selection within that cohort does not remove the original coverage limits; purposive restrictions also shape the scope of the claims.

13. How much evidence is enough?

Should the quantitative and qualitative sample sizes match? Justify each against its task.

Evidence needed	Plan and justify	Check the inference
Quantitative strand	Precision or power for the planned analysis; variation, clustering, non-response, and attrition.	Which population or cohort can the results describe?
Qualitative strand	Information richness, relevant variation, depth of engagement, and adequacy for the chosen analysis.	Which experiences and contexts are represented or missing?
Integration	Enough linked cases or comparable coverage to examine the relationship between findings.	Can unmatched, missing, or contrasting cases change the combined explanation?

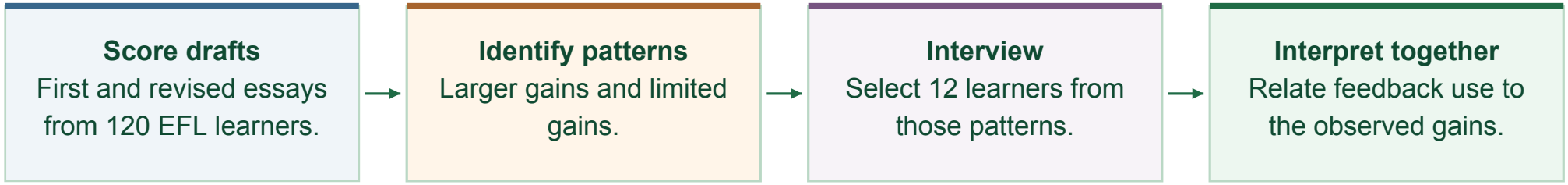
Our 120 scored learners and 12 interviewees are illustrative choices. Justify the size of each actual study sample and the evidence linking them.

Write the sampling plan before collecting data

Specify the target population and units, selection rules for each strand, sample relationship, timing, size rationale, consent for linkage or recontact, and how losses or changes will be reported.

14. Follow one route: explain revision gains

For our running study, we choose explanatory sequential design: first establish the revision pattern, then investigate how learners used the feedback.



The interview phase depends on the first-phase results; it is not just a second method added later.

15. Who can explain the pattern?

Decide who to interview

Scores from 120 learners show 48 larger gains and 72 limited gains. Which selection best helps explain that difference?

A. Interview only the six learners with the highest scores.

B. Purposefully select six learners from each gain pattern.

C. Interview twelve volunteers without examining their gain patterns.

Choose an answer and explain why before checking the following page.

Who can explain the pattern? - worked interpretation

Suggested answer: B. This connects the interview sample to both initial patterns. The balanced illustrative selection supports comparison; it does not represent the 40%/60% distribution.

Inspect the sampling plan and its limits

Illustrative result	Purposeful selection	What to understand
48 of 120 improve by at least one point.	Select six learners from this pattern.	How were comments used to develop an argument?
72 of 120 improve by less than one point.	Select six learners from this pattern.	What was difficult to understand or apply?

Scores refer to argument development on a 0–4 rubric. The one-point split is an illustrative decision, not a universal sampling rule.

Obtain consent for recontact, retain secure participant links, and report who declines or cannot be reached.

16. Let the results shape the interviews

Choose a useful follow-up question

A learner shows limited argument-development gains. Which interview question most directly investigates this result?

A. Did you like the teacher’s feedback?

B. Why did the feedback fail to improve your argument?

C. Looking at this comment and your draft, what did you try to change, and what was difficult?

Choose an answer and explain why before checking the following page.

Let the results shape the interviews - worked interpretation

Suggested answer: C. This connects the initial pattern to a specific comment, an attempted revision, and the learner's explanation without assuming a cause.

Inspect the full interview bridge

Quantitative pattern	Interview focus	Evidence to examine together
Larger gains	How did you decide what evidence to add or explain?	Teacher comment → draft change → student explanation.
Limited gains	Which comment was hardest to use, and what did you try?	The same comment and the related draft passage.
Unexpected case	Why does the draft change differ from what you described?	Revisit the score, the passage, and the interview interpretation.

Sampling connects the phases. The results also help build focused interview questions.

17. Make the connection visible

Operation	What it does	Education example
Connecting	Links the samples.	Draft-score patterns guide interview selection.
Building	Shapes the next data collection.	Interview findings inform feedback-literacy items.
Merging	Brings findings together for comparison.	Compare revision scores and interview findings by writing issue.
Embedding	Links inquiry at several points in a larger study.	Explore needs before an AI-feedback intervention and examine implementation during it.

Our chosen route connects sampling and builds the interviews, then brings the linked findings together for interpretation.

Read the integration methods article

18. Two sets of findings are ready

Evidence	Illustrative finding
Draft scores: 120 learners	48 improve by at least one point; 72 improve by less than one point on argument development.
Interviews: 12 purposefully selected learners	Some describe using comments to strengthen claim–evidence links. Others mainly describe correcting language or difficulty planning a substantive revision.

Reporting these in separate paragraphs is only the starting point. What does reading them together add?

19. Build an integrated explanation

Illustrative linked dataset: 120 scored draft pairs; 12 interviews. The findings below are invented.

Revision pattern	Interview finding	Integrated interpretation
Larger gains: 48/120 (40%)	Selected students describe adding evidence and explaining why it supports a claim.	What would you infer?
Limited gains: 72/120 (60%)	Selected students describe grammar-focused edits or uncertainty about how to revise the argument.	What would you infer?

Complete the interpretation

Selected learners with limited gains describe grammar edits or difficulty planning an argument revision. What does linking these findings add?

A. Understanding a comment may differ from knowing how to use it to revise an argument.

B. Sixty per cent of all learners lacked revision-planning skills.

C. The interviews prove that grammar correction caused the limited gains.

Choose an answer and explain why before checking the following page.

Build an integrated explanation - worked interpretation

Suggested answer: A. This offers a tentative explanation connecting what the scores measure with what selected learners describe. Check it against linked drafts and other cases.

Inspect a completed display and its limits

Revision pattern	Interview finding	Integrated interpretation
Larger gains: 48/120 (40%)	Selected students describe adding evidence and explaining why it supports a claim.	Their explanations suggest how acting on argument-focused feedback may relate to larger gains.
Limited gains: 72/120 (60%)	Selected students describe grammar-focused edits or uncertainty about how to revise the argument.	Understanding what a comment means may differ from knowing how to use it in a revision.

An integrated inference states what the relationship adds. Interviews with 12 learners do not establish how common each explanation is.

20. What if the evidence does not agree?

A student’s argument score improves, but the interview mainly describes grammar correction.

Check comparability
Same task, passage, time, and scoring focus?

→

Revisit evidence
Inspect the draft changes, scoring, and interview interpretation.

→

Report the difference
Explain what remains uncertain; follow up if justified.

A difference may reveal incomplete recall, a scoring issue, or another route to improvement. Do not force all cases into one explanation.

21. What can we claim?

Choose a defensible conclusion

We have draft scores from 120 learners and linked interviews with 12 selected learners. Which conclusion fits this evidence?

A. Teacher feedback caused the improvements in this cohort.

B. All learners with limited gains misunderstood the feedback.

C. Differences in how feedback was used may help explain revision gains in this cohort.

Choose an answer and explain why before checking the following page.

What can we claim? - worked interpretation

Suggested answer: C. This preserves the integrated contribution while keeping the explanation tentative and bounded to this study.

Inspect the conclusion and its limits

Differences in how feedback was interpreted and used may help explain revision gains in this cohort.

- The score pattern describes change; it does not prove that feedback caused the change.
- The interviews support possible explanations for selected learners; they do not estimate their prevalence.
- Check measurement, qualitative analysis, missing cases, sample links, and alternative interpretations.

The integrated explanation must be supported by both strands and by the way they were linked.

22. What must the reader be able to follow?

Section	Keep the research story visible
Introduction	Why revision gains need both quantitative and qualitative evidence.
Method	Which mixed design, which samples, which strand analyses, and exactly how the strands were linked.
Results	The score pattern, the qualitative findings, and the integrated explanation.
Discussion	What the integration adds, what differs, and what remains uncertain.

Download JARS–Mixed (PDF)

JARS–Qual

JARS–Quant

Use the mixed-methods standard together with the relevant qualitative and quantitative reporting standards.

23. Make the connection traceable in the Method

Model paragraph for the invented teaching study

We used an explanatory sequential mixed methods design to investigate differences in EFL essay revision. We scored argument development in first and revised drafts from 120 learners using a 0–4 rubric.

The initial gain patterns guided purposeful selection of 12 learners and the interview questions. Interviews focused on how learners interpreted and used specific comments, and were examined through qualitative content analysis.

We linked the gain patterns and qualitative findings in a joint display to develop explanations of the observed differences.

A complete Method must also justify the rubric, sampling criteria, analytic procedures, researcher role, and ethical decisions.

24. Report what the integration adds

Model paragraph for the same invented dataset

Of 120 learners, 48 improved argument development by at least one point and 72 improved by less than one point. Interviews with 12 selected learners described differences in how they used teacher feedback.

The joint display suggests that applying comments to claim–evidence links may help explain larger gains, while difficulty planning substantive revisions may help explain limited gains. These explanations are tentative and do not establish prevalence or causation.

Evidence → relationship between findings → added explanation → limits

25. Keep the combined finding in the abstract

Move	Our EFL feedback example
Purpose	Explain differences in essay revision after teacher feedback.
Design and sources	Explanatory sequential; scored drafts followed by linked interviews.
Analysis and integration	Describe gains; analyse interview content; connect findings in a joint display.
Integrated finding	Different ways of using feedback may help explain different gains.
Scope	An interpretation of this cohort, with sampling and causal limits.

Follow the journal’s abstract structure and word limit. Keep the integrated contribution visible.

26. Is the integration actually reported?

We used mixed methods. Essay scores were followed by interviews. The interviews confirmed the results and improved validity.

Repair the reporting

Which addition would make the integration claim most traceable?

A. “Using two methods provided triangulation and increased validity.”

B. “Gain patterns guided interview selection and questions; a linked display compared draft changes with interview findings, including different cases.”

C. “A larger sample was used, and both datasets were analysed carefully.”

Choose an answer and explain why before checking the following page.

Is the integration actually reported? - worked interpretation

Suggested answer: B. This identifies what linked the phases and where findings were compared. The report must also show the actual evidence and explain the resulting inference.

What would you need to see before accepting these claims?

Check the missing evidence

- Did score patterns shape interview sampling and questions?
- Which participants and writing issues link the strands?
- Where is the evidence supporting the integrated interpretation?
- Were different findings and other explanations examined?
- Which specific quality problem was addressed?

27. Review an existing mixed-methods study

Use an article from your field, or the open-access ESL writing study below. Work from its Method and Results, not its label alone.

Open Mahapatra (2024): ChatGPT formative feedback in ESL writing

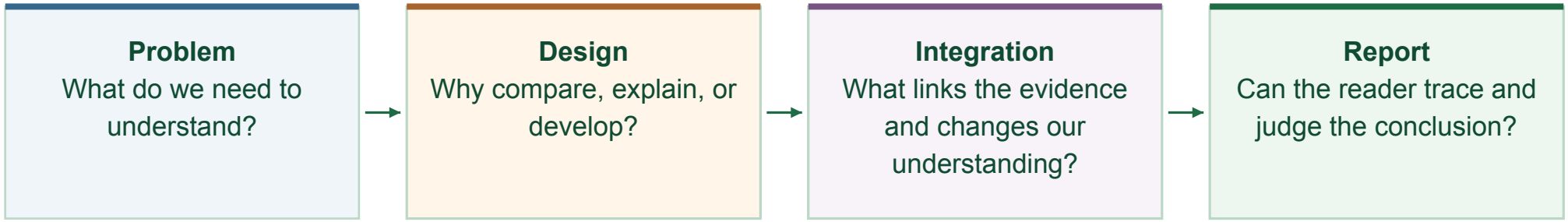
Download the article (PDF)

- Find the design diagram and identify the purpose of each strand.
- Trace timing, participant links, and the integration procedure.
- Point to a passage, table, or figure showing what integration adds.
- Judge which claims the evidence supports and what remains unclear.

Evidence to locate in the provided article

The reported quantitative sample includes 35 intervention and 37 comparison students. Five intervention students participated in all three focus groups. Examine the sample link, attrition, and the integrated interpretation; the design label alone does not establish quality.

28. Follow the research story



For the EFL feedback study: score patterns identified a difference; linked interviews helped develop an explanation; the report made the link and its limits visible.

29. Thank you

Questions and discussion

What would integrating two strands add to a study you are reading?

Open the interactive lesson